Design Verification Plan & Report

VBF-T4R2-DVPR-01 | case t4_r2
Case t4_r2 · finalist E3-porheadroom · DFN verdict grade · 2026-08-26

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> One row per verification item. Values mechanically taken from simulation outputs under cell/; uncovered conditions honestly marked "N/A (requires physical experiment)". No values from memory.

Conclusion

All five entry-0 criteria PASS at DFN verdict grade with margins (retention +4.6 ppt; +132.3 Wh/kg; +56.8 Wh/L; −6.19 K thermal; plating margin 6 mV). Two independent DFN 4C runs agree exactly.

Uncovered (cited for paper limitations): physical abuse (nail/crush/drop), overcharge-to-runaway calibration, cycle life, sub-−20 °C operation, and any molecular-level true-compute endorsement (Stage 5 skipped, real_compute=false). The thinnest margin is the plating headroom (6 mV) — flagged as the top DFMEA risk with mitigations in dfmea.md.

Item	Condition	Result value	Determination	Source
1C retention @ −20 °C (W1)	lowT_discharge, T_0=253.15 K cold-soak (warm-start artifact procedure), DFN	0.9938 (5.0078 / 5.0392 Ah)	✓ PASS vs ≥ 0.95 (margin +4.6 ppt)	cell/r5_E3_retention_dfn.json
Gravimetric energy density (W2)	1C discharge energy ÷ contract-caliber mass (electrolyte excluded)	459.4465 Wh/kg	✓ PASS vs ≥ 327.18 (margin +132.3)	cell/r5_E3_energy_dfn.json
Volumetric energy density (W3)	contract caliber: Σ layer thickness × area	936.8218 Wh/L	✓ PASS vs ≥ 880 (margin +56.8)	cell/r5_E3_energy_dfn.json
4C fast-charge temperature rise (W4)	4C_charge_45C (pre-condition 1C discharge to 2.5 V then 4C charge to 4.2 V, ambient 318.15 K), DFN + lumped thermal	T_max 326.96 K	✓ PASS vs ≤ 333.15 (margin −6.19 K)	cell/r5_E3_4c_dfn2.json
4C fast-charge plating (W5)	same protocol + plating module, anode_potential_v vs 0 V	min +0.0060 V, 0/307 points negative → plated=false	✓ PASS (margin 6 mV)	cell/r5_E3_4c_dfn2.json
Reproducibility	second independent DFN 4C run	identical T_max and anode-potential min	✓ deterministic	r5_E3_4c_dfn.json vs _dfn2
1C discharge capacity (25 °C)	1C_discharge, DFN	5.0392 Ah	informational (no capacity criterion; +0.78 % vs nominal 5.0)	cell/r5_E3_1c_dfn.json
Voltage window	parameter set limits	2.5 – 4.2 V	informational	Chen2020 parameter set
Nail penetration	—	N/A (beyond pure simulation boundary, requires physical experiment)	—	—
Overcharge to thermal runaway	run-tr module exists but no abuse criterion in entry-0 stage set	N/A (requires physical experiment / not in scope)	—	—
Crush / drop	—	N/A (requires physical experiment)	—	—
Cycle life	aging protocol exists but not in entry-0 scope	N/A — "Not simulated (no aging run)" — not fabricated	—	—
Rate-pulse internal resistance	not a protocol	N/A (DCR 2.07 mΩ reported in calc-energy as informational)	—	r5_E3_energy_dfn.json dcr_ohm